Feasibility Analysis of In-Band Coexistence in Dense LEO Satellite Communication Systems

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***Abstract*— This work provides a rigorous assessment of the feasibility of spectrum sharing between large low-earth orbit (LEO) satellite constellations. For concreteness, we focus on the existing Starlink system and the soon-to-be-launched Kuiper system, the latter of which is prohibited from inflicting excessive interference onto incumbent Starlink ground users. We carefully model and study the potential downlink interference between the two systems at 20 GHz and investigate how strategic satellite selection may be used by Kuiper to serve its own ground users while also protecting Starlink ground users. We then extend this notion of satellite selection to the case where Kuiper has limited knowledge of Starlink's serving satellite. Throughout our analysis, we examine the distribution of interference and SINR over time as each constellation orbits the globe. Our findings reveal that there is virtually always the potential for very high or extremely low interference, depending on which Starlink and Kuiper satellites are being used to serve their ground users. Consequently, we show that Kuiper can protect Starlink ground users, with high probability, by strategically selecting which of its satellites are used to serve its ground users. Simultaneously, Kuiper is capable of delivering near-maximal downlink SINR to its own ground users. This highlights a potential feasible route to the coexistence of two dense LEO satellite systems, even in scenarios where one system has limited knowledge of the other's serving satellites.**

***Index Terms*— Dense low earth orbit (LEO) satellites, satellite communications, 6G, satellite system coexistence, mega constellations.**

I. Introduction

A NEW paradigm of global broadband connectivity is unfolding as mega-constellations comprised of thousands of LEO satellites are deployed to deliver wireless coverage across the globe. Although they face a multitude of noteworthy challenges, these LEO satellite communication systems circumvent the time-consuming obstacles associated with deploying terrestrial infrastructure, which has left many regions and communities severely under-served—or even completely unserved—by terrestrial-based modes of connectivity [1]. Two of the most notable large-scale efforts in this pursuit are SpaceX's Starlink [2], with over 5,000 satellites currently in orbit, and Amazon's Project Kuiper [3], which recently launched test satellites in October 2023 and intends to launch 3,236 satellites throughout the decade [4].

These mega-constellations have already transformed the connectivity landscape and will be a valuable new source of broadband access in the 6G era [5], [6], [7], [8]. Unlike in traditional terrestrial cellular networks, frequency spectrum has been allocated to these satellite systems in a *non-exclusive* manner by the Federal Communications Commission (FCC) in the United States and other global spectrum regulators [9], [10]. To facilitate fair spectrum sharing, the FCC gives precedence (or incumbency) to systems which applied for launch rights in earlier so-called processing rounds than others. Consequently, it is the expectation of the FCC that each system either coordinates with or protects systems which acquired launch rights in earlier processing rounds [11].

Given the sheer number of satellites slated to launch and the growing number requesting launch rights, spectrum sharing across these LEO satellite systems is far from trivial, especially when one considers the dynamics over time as satellites orbit. The feasibility of this coexistence is currently unclear and mechanisms to facilitate coexistence are even less established [11]. In this work, we take a necessary first step by carefully investigating the severity of in-band interference over time between coexisting LEO satellite systems as their constellations orbit the globe, which ultimately sheds light on the efficacy of potential mechanisms to protect incumbent systems—and how imposing this protection may impact system performance.

A. Related Work

The principle of spectrum sharing is concerned with so-called *secondary* systems not inflicting significant interference onto *primary* (or incumbent) systems when attempting to access some portion of frequency spectrum [12], [13], [14]. Various mechanisms have been studied to mitigate interference in spectrum sharing [15], [16], [17]. In particular, spatial domain beamforming employing an *avoidance angle* or *look-aside* technique imposes a minimum separation angle between a primary satellite-user link and interfering links from secondary systems [18]. These techniques were initially proposed to protect geostationary orbit (GSO) systems from harmful interference caused by non-geostationary orbit (NGSO)

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systems but also have promise in mitigating interference between NGSO satellites [16], [19], [20], including LEO and medium-earth orbit (MEO) satellites. The work [21] and [22] concludes that downlink interference of early-stage NGSO satellite communication systems of the 1990s is effectively managed by high beam gain antennas and frequency reuse. A notable survey of existing work on interference studies of modern satellite systems [19] draws a similar conclusion that interference between satellite systems is well controlled by a high beam gain mask and an avoidance angle of at least $5°$, but this conclusion should be re-evaluated in the context of emerging dense LEO satellite systems. The work of [23] uses stochastic geometry to investigate downlink interference and coverage probability in terms of signal-to-interference-plus-noise ratio (SINR), assuming mutual satellite positions to be random processes. This work does not strictly capture how interference varies as a function of beams' steering directions and does not account for any explicit interference mitigation mechanisms.

The authors in [16] and [20] conduct extensive assessments of the user link interference between NGSO systems in terms of throughput degradation, applying various interference mitigation techniques such as look-aside and band-splitting. The primary and secondary systems were assumed to randomly select their serving satellites from those above a certain elevation angle, with the constraint that the secondary system's serving satellite was subject to a minimum specified angular separation between its satellite-user link and the primary satellite-user link. Both studies [16], [20] acknowledge that the look-aside helps decrease interference to some extent but has limited benefits for small constellations and is inefficient under heterogeneous orbits (HEO).

While the studies of [16] and [20] are insightful, they do not provide a clear picture of the feasibility of spectrum sharing among non-cooperative mega-constellations of LEO satellites and do not explore more sophisticated approaches to secondary satellite selection beyond look-aside which guarantee some level of both downlink quality and protection. Open questions remain regarding the number of secondary satellites which offer sufficiently low interference under techniques akin to look-aside, the bounds on interference between systems, the prospects of secondary satellite selection to enable coexistence with incomplete knowledge of the primary system, and how all of this varies as the satellites orbit the globe over time.

B. Contributions

In this paper,[1] we analyze the coexistence paradigm laid forth thus far: the operation of a *primary* (or incumbent) LEO satellite system in the presence of a *secondary* LEO satellite system attempting to coexist in-band under the expectation that it protects the primary system. In light of active discussions in the regulatory domain and the impending launch of thousands of additional LEO satellites, this paper investigates the feasibility of strategic satellite selection by the secondary system to serve its own ground user while simultaneously protecting a co-located primary ground user from excessive interference. In this pursuit, we examine bounds on downlink interference (satellite to ground user) and its statistics over time and across satellites in orbit. Further, we investigate the number of secondary satellites capable of protecting a primary ground user, inflicting onto it interference below some threshold. This then motivates us to explore the feasibility of strategically selecting a secondary satellite to maximize its downlink service while maintaining protection.

We accomplish this by creating a high-fidelity simulation of two LEO satellite systems—namely Starlink and Kuiper—using actual orbital parameters, transmit powers, antenna gains, and other system parameters reported in public FCC filings [25], [26]. Our aim is to shed light on the feasibility of in-band coexistence in LEO satellite communication systems and to motivate future work proposing mechanisms to enable such coexistence. To accomplish this, we broadly investigate three important questions surrounding this coexistence.

How much interference will be inflicted by the secondary system onto the primary? Perhaps the most fundamental concern we explore in this work is quantifying how much in-band interference a secondary satellite may inflict onto a primary ground user. To do this, in Section IV, we evaluate the absolute bounds on this interference, as well as the bounds when conditioning on a particular primary satellite serving the ground user. We examine the distributions of both of these bounds over time as the satellites orbit the globe and conclude that interference can be very high or extremely low at virtually any given time, depending heavily on the pair of primary and secondary satellites being used to serve downlink.

Can the secondary system protect the primary and what does it sacrifice in doing so? In Section V, we then investigate how feasible it is for the secondary system to reliably protect a primary user under various interference constraints. In addition to mere feasibility, we also examine the number of secondary satellites capable of meeting this constraint. We then illustrate that strategic satellite selection may be used by the secondary system to serve its own ground user while also protecting a primary ground user. In doing so, we dissect the sacrifices in secondary system performance under such a technique, in terms of received SINR. Again, we show all of this in distribution across the constellations' orbits and remarkably discover that the secondary system often only sacrifices a fraction of a decibel in SINR while guaranteeing protection.

How much knowledge does the secondary system need about the primary? Finally, in Section VI, we conclude by reexamining the viability of protection when the secondary system has limited knowledge about which primary satellite is serving a given ground user. We observe that, on average, there are 10 secondary satellites capable of guaranteeing that primary ground users are protected, even when the secondary system has only modest knowledge about the vicinity of the primary serving satellite. While more uncertainty and a more stringent protection constraint certainly make it more difficult to guarantee protection, we impressively find that the secondary system only sacrifices at most around 2 dB of SINR in median and around 4 dB in the worst case. We nonetheless highlight that minimal coordination between the two systems has the potential to close this gap.

[1]This work is an extended version of [24].

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II. System Model

This work considers two downlink LEO satellite communication systems, each operating independently to serve corresponding ground users on the surface of the Earth. Each of the two systems is comprised of a unique constellation of satellites orbiting at altitudes on the order of 500 km. We term one of the systems the *primary* system and the other the *secondary* system. A dense LEO satellite system generally intends to provide global service coverage using thousands of satellites in orbit, each equipped with multiple spot beams, by systematically steering spot beams toward their corresponding ground cells. There can therefore be significant overlap in the coverage areas of both systems. We consider Earth-fixed ground cells, meaning that ground cells remain stationary while the beams directed at them change as satellites orbit the Earth.

The work herein is interested in assessing the in-band interference inflicted onto a given ground user of one system by a single satellite of the other system. When assessing the feasibility of coexistence and interference protection, the key factor is whether the interference of any primary ground user exceeds an established threshold. Our focus is therefore directed to worst-case interference inflicted by the secondary system, which occurs when primary and secondary ground users are co-located and the secondary satellite's beam is steered directly toward its user. As such, we consider the case where the two systems are transmitting downlink to nearby ground users at the same time and with the same carrier frequency of 20 GHz. This corresponds to scenarios where the secondary system may intentionally or unintentionally transmit downlink using the same frequency resources as a primary system. In both systems, we assume each satellite employs high-gain antennas to form a highly directional beam in the general direction of the ground user it aims to serve. While additional interference may arise from other spot beams in the secondary system in practice, we narrow our focus to the interference inflicted onto a primary user by the serving satellite of a co-located secondary user, as this is the dominant source of interference and thus has the greatest impact on coexistence. This is especially true given that the secondary system will presumably carefully manage the interference it inflicts on itself across spot beams and given that practical systems time- and frequency-multiplex spot beams, providing substantial spatial separation (and thus isolation) between highly directional spot beams across the constellation. Nonetheless, we would like to remark that any additional interference introduced could be directly accounted for in this work via the interference protection threshold we will introduce shortly.

Let us define the transmit antenna gain of the primary satellite $\mathbf{p}$ in the direction of a ground user $\mathtt{u}$ as $G_{\mathrm{tx}}(\mathtt{u},\mathbf{p})$. Similarly, let $G_{\mathrm{rx}}(\mathtt{u},\mathbf{p})$ be the receive antenna gain of user $\mathtt{u}$ in the direction of primary satellite $\mathbf{p}$. The transmit or receive gain pattern is determined by the antennas used on satellites or by ground users. In the following section, we will introduce the specific beam gain patterns under consideration, taking into account regulatory requirements. The path loss between the two we denote as $L(\mathtt{u},\mathbf{p})$. With this, we can define the received signal-to-noise ratio (SNR) at a primary ground user $\mathtt{u}$ served by its serving satellite $\mathbf{p}$ as

$$\mathsf{SNR}(\mathtt{u},\mathbf{p}) = \frac{P_{\mathrm{tx}}(\mathbf{p})\ G_{\mathrm{tx}}(\mathtt{u},\mathbf{p})\ G_{\mathrm{rx}}(\mathtt{u},\mathbf{p})\ L(\mathtt{u},\mathbf{p})^{-1}}{P_{\mathrm{n}}(\mathtt{u})}, \quad (1)$$

where $P_{\mathrm{tx}}(\mathbf{p})$ is the transmit power of $\mathbf{p}$ and $P_{\mathrm{n}}(\mathtt{u})$ is the noise power at $\mathtt{u}$. We assume line-of-sight (LOS) channels based on conclusions in [27], a 3GPP report which found this to be the case the vast majority of the time in outdoor environments for satellites at the elevations under consideration. The path loss $L(\mathtt{u},\mathbf{p})$ from a satellite $\mathbf{p}$ to a rooftop-mounted ground user $\mathtt{u}$ is modeled based on free-space path loss as atmospheric and scintillation losses in the mid-latitude regions at elevation angles above 35° are often observed to be marginal [27], [28], [29], and shadowing is negligible under clear sky LOS condition [30]. Also, clutter loss which accounts for the attenuation of signal power caused by surrounding buildings and objects on the ground is also negligible when the user is in an LOS condition [27]. Thus, the path loss $L(\mathtt{u},\mathbf{p})$ we consider is

$$[L(\mathtt{u},\mathbf{p})]_{\mathrm{dB}} = 32.45 + 20\log_{10}(f_c) + 20\log_{10}(d(\mathtt{u},\mathbf{p})), \quad (2)$$

where f_c is the carrier frequency of 20 GHz and $d(\mathtt{u},\mathbf{p})$ is the path distance from the ground user $\mathtt{u}$ and the satellite $\mathbf{p}$. Generalizing the notation to the secondary system, the received SNR at a secondary ground user $\mathtt{v}$ from its satellite $\mathbf{s}$ is

$$\mathsf{SNR}(\mathtt{v},\mathbf{s}) = \frac{P_{\mathrm{tx}}(\mathbf{s})\ G_{\mathrm{tx}}(\mathtt{v},\mathbf{s})\ G_{\mathrm{rx}}(\mathtt{v},\mathbf{s})\ L(\mathtt{v},\mathbf{s})^{-1}}{P_{\mathrm{n}}(\mathtt{v})}. \quad (3)$$

In this paper, the notation $\mathbf{p}$ and $\mathbf{s}$ are used to identify unique primary and secondary satellites and their locations; similarly, $\mathtt{u}$ and $\mathtt{v}$ are used to identify the primary and secondary ground users and their locations.

As mentioned, we are particularly interested in the interference inflicted onto each system by the other. In this work, our focus is exclusively on the in-band interference inflicted by individual satellites from each system, in order to draw definitive conclusions on the interplay between two satellites, which, to a certain extent, may be extrapolated to sets of satellites. For concreteness, this work does not consider other sources of interference, such as from other beams or satellites across the constellations. Extending the analysis herein to such is undoubtedly a worthwhile direction for future work but would hinge on other factors involving satellite association policies and handover.

Suppose a secondary satellite $\mathbf{s}$ inflicts interference onto a primary ground user $\mathtt{u}$ being served by a primary satellite $\mathbf{p}$. Then, the resulting interference-to-noise ratio (INR) at the primary ground user $\mathtt{u}$ is

$$\mathsf{INR}(\mathtt{u},\mathbf{p};\mathbf{s}) = \frac{P_{\mathrm{tx}}(\mathbf{s})\ G_{\mathrm{tx}}(\mathtt{u},\mathbf{s};\mathtt{v})\ G_{\mathrm{rx}}(\mathtt{u},\mathbf{s};\mathbf{p})\ L(\mathtt{u},\mathbf{s})^{-1}}{P_{\mathrm{n}}(\mathtt{u})}. \quad (4)$$

Here, we have slightly extended the notation of $G_{\mathrm{tx}}(\cdot)$ and $G_{\mathrm{rx}}(\cdot)$ to make it clear that $G_{\mathrm{tx}}(\mathtt{u},\mathbf{s};\mathtt{v})$ represents the transmit gain of the secondary satellite $\mathbf{s}$ in the direction of the primary user $\mathtt{u}$ when $\mathbf{s}$ serves its secondary ground user $\mathtt{v}$. Similarly,

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$G_{\mathrm{rx}}(\mathrm{u}, \mathbf{s}; \mathbf{p})$ represents the receive gain in the direction of the secondary satellite $\mathbf{s}$ when the ground user u steers its antenna toward its serving satellite $\mathbf{p}$. By virtue of this, the primary ground user's INR depends on its primary serving satellite $\mathbf{p}$ and on the secondary satellite $\mathbf{s}$. A primary satellite $\mathbf{p}$ likewise will inflict interference onto a secondary ground user v being served by satellite $\mathbf{s}$, leading to an INR at v of

$$\mathsf{INR}(\mathrm{v}, \mathbf{s}; \mathbf{p}) = \frac{P_{\mathrm{tx}}(\mathbf{p})\ G_{\mathrm{tx}}(\mathrm{v}, \mathbf{p}; \mathrm{u})\ G_{\mathrm{rx}}(\mathrm{v}, \mathbf{p}; \mathbf{s})\ L(\mathrm{v}, \mathbf{p})^{-1}}{P_{\mathrm{n}}(\mathrm{v})}. \tag{5}$$

Together, the received SNR and INR at the primary ground user u and at the secondary ground user v dictate their SINRs, which take the familiar forms

$$\mathsf{SINR}(\mathrm{u}, \mathbf{p}; \mathbf{s}) = \frac{\mathsf{SNR}(\mathrm{u}, \mathbf{p})}{1 + \mathsf{INR}(\mathrm{u}, \mathbf{p}; \mathbf{s})}, \tag{6}$$

$$\mathsf{SINR}(\mathrm{v}, \mathbf{s}; \mathbf{p}) = \frac{\mathsf{SNR}(\mathrm{v}, \mathbf{s})}{1 + \mathsf{INR}(\mathrm{v}, \mathbf{s}; \mathbf{p})}. \tag{7}$$

A prohibitively high INR (e.g., $\mathsf{INR} \gg 0$ dB) would lead to significant degradation in link quality where $\mathsf{SINR} \ll \mathsf{SNR}$. As such, we are motivated to investigate the severity of interference in the sequel, but first we lay the groundwork on the methodology by which we accomplish such an investigation.

III. Methodology of Our Feasibility Analysis

To conduct a thorough analysis on the feasibility of two dense LEO satellite communication systems coexisting in-band at 20 GHz, we consider the two preeminent commercial systems mentioned in the introduction: Starlink by SpaceX and Project Kuiper by Amazon. We consider Starlink as the primary system and Kuiper as the secondary system; this is motivated by the fact that Starlink has priority rights to the 19.7–20.2 GHz band for downlink transmission [11]. Indeed, based on current regulations, Kuiper is permitted to operate within that band along with Starlink, with the understanding that it will not cause prohibitive interference to Starlink [3]—coinciding with the analyses herein.

A. LEO Satellite Constellations

We simulate the constellation of satellites in Starlink and Kuiper in a Walker-Delta fashion [31], realized using MATLAB's Aerospace Toolbox [32], based on the orbital parameters detailed in Table I and Table II, which are extracted from public filings [2], [3].[2] Both systems' constellations are comprised of multiple orbital shells at various altitudes; each shell is independently realized using the Walker-Delta model. With this model, satellites in a given shell are deployed in multiple equal-radius (i.e., altitude) circular orbits, with the same number of satellites per orbital plane, where each orbital plane has an identical inclination to a reference plane [31].[3]

Let $\bar{\mathcal{P}} = \{\mathbf{p}_i : i = 1, \ldots, 4408\}$ be the set of all 4408 satellites in Starlink's constellation. Similarly, let $\bar{\mathcal{S}} = \{\mathbf{s}_i : i = 1, \ldots, 3236\}$ be the set of all 3236 satellites in Kuiper's constellation. At a given instant, the systems aim to serve some primary ground user u and secondary ground user v.

[2]This work considers the first-generation Starlink constellation.

[3]For orbital shells with the same altitude and inclination in Table I, we adjust the right ascension of the ascending node (RAAN) of these shells to ensure that any planes from each other's shells do not overlap.

TABLE I

SpaceX's Starlink Constellation Parameters [2]

Altitude	Inclination	Planes	Satellites/Plane	Total No. Satellites
540 km	53.2°	72	22	1584
550 km	53°	72	22	1584
560 km	97.6°	4	43	172
560 km	97.6°	6	58	348
570 km	70°	36	20	720

TABLE II

Amazon's Project Kuiper Constellation Parameters [3]

Altitude	Inclination	Planes	Satellites/Plane	Total No. Satellites
590 km	33°	28	28	784
610 km	42°	36	36	1296
630 km	51.9°	34	34	1156

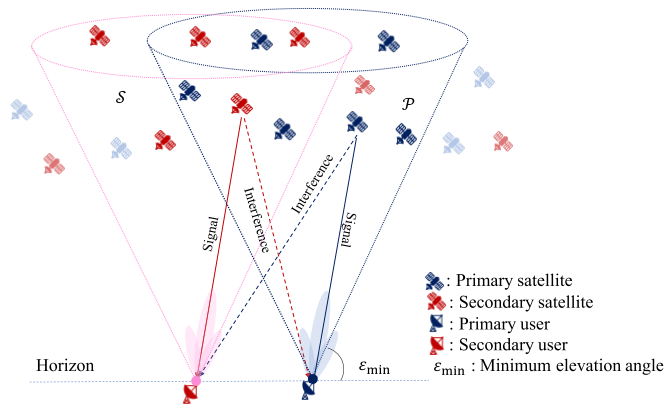

Fig. 1. This work considers a scenario where a primary satellite system and a secondary satellite system interfere with one another when attempting to serve downlink to ground users at the same time and at the same frequency.

When aiming to serve these users in practice, each of these systems will select one satellite within its set of *overhead* satellites. More specifically, each system will select a satellite that is within some cone centered on the vector normal to the surface of the Earth at the ground user's location, as shown in Fig. 1. In practice, this set of overhead satellites is typically defined by a minimum elevation angle $\epsilon_{\min} > 0$ relative to the user's horizon, at or above which a satellite must lie to be considered for selection. It is reasonable to assume both systems employ the same minimum elevation angle, which we assume herein to form the sets of overhead satellites as

$$\mathcal{P} = \left\{\mathbf{p} \in \bar{\mathcal{P}} : \epsilon(\mathrm{u}, \mathbf{p}) \geq \epsilon_{\min}\right\} \subset \bar{\mathcal{P}}, \tag{8}$$

$$\mathcal{S} = \left\{\mathbf{s} \in \bar{\mathcal{S}} : \epsilon(\mathrm{v}, \mathbf{s}) \geq \epsilon_{\min}\right\} \subset \bar{\mathcal{S}}, \tag{9}$$

where $\epsilon(\mathrm{u}, \mathbf{p})$ denotes the elevation of a satellite $\mathbf{p}$ relative to the horizon at a user u; we take $\epsilon_{\min} = 35°$ [3][4] in the results that follow.

While $\bar{\mathcal{P}}$ and $\bar{\mathcal{S}}$ do not change over time, as they represent the set of all primary and secondary satellites in orbit, both

[4]Per filings, $\epsilon_{\min} = 40°$ for Starlink [2] and $\epsilon_{\min} = 35°$ for Kuiper [3] when serving ground users.

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TABLE III

KEY SIMULATION PARAMETERS [25], [26], [35]

Satellites				
Tx. Antennas	No. Elements	64×64		
	3 dB Beamwidth	1.6°		
	Max. Tx. Gain	36 dB		
Max. EIRP	Primary Sat.	−54.3 dBW/Hz		
	Secondary Sat.	−53.3 dBW/Hz		
Ground Users				
Rx. Antennas	No. Elements	8×8	16×16	32×32
	Max. Rx. Gain	18 dB	24 dB	30 dB
	3 dB Beamwidth	12.8°	6.4°	3.2°
Noise Power Spectral Density		−174 dBm/Hz		
Noise Figure		1.2 dB		

$\mathcal{P}$ and $\mathcal{S}$ are functions of time, since the satellites overhead a particular user will vary as the constellations progress in their orbit; in LEO systems, satellites are overhead a typical user for about two minutes at most [5], [33]. However, we omit explicit time indexing for brevity and clarity in notation. To capture this time dimension, we simulate the system over 24 hours with a resolution of 30 seconds. In other words, every 30 seconds, we observe a snapshot of the constellations to obtain satellite locations, simulate interference, and conduct our analysis. This process is repeated nearly 3,000 times over a 24 hour period as satellites progress their orbits. Then, we present metrics such as INR and SINR as their empirical cumulative distribution functions (CDFs) across approximately 3,000 different ground users per system, taken over this 24 hour period.

B. Satellites and Ground Users

Having considered the 19.7–20.2 GHz band, we simulate the system at a carrier frequency of 20 GHz with 400 MHz of bandwidth. We assume all satellites and ground users in both the primary and secondary systems are equipped with standard uniform planar antenna arrays with half-wavelength spacing. We assume satellites are equipped with 64×64 antenna arrays containing 4096 elements and perform canonical matched filter beamforming toward the user they are serving[5]; this delivers a maximum gain that is on par with the 34.5–37 dBi reported in filings [25], [26]. The transmit power of each primary satellite is set to a maximum effective isotropic radiated power (EIRP) of −54.3 dBW/Hz and that of each secondary satellite is set to −53.3 dBW/Hz, according to FCC filings of Starlink [25] and Kuiper [26] when operating in the 20 GHz band. The key simulation parameters are summarized in Table III. In each constellation, we assume satellites orbiting at the highest altitude transmit with maximum EIRP. For satellites at lower altitudes, the EIRP is reduced slightly to account for differences in path loss to ensure the received signal strength from satellites orbiting at various altitudes are approximately equal. This is on par with power control techniques employed in practical systems, which must abide by power flux density constraints imposed by regulators and thus cannot always transmit with maximum EIRP [25], [26], [36].

[5]For per-cell beamforming, users can be assumed located at the cell centers.

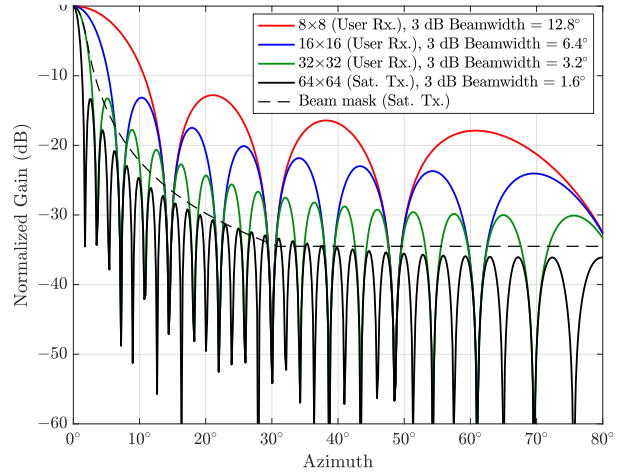

Fig. 2. Beam patterns of the 64×64 phased array antenna considered at each satellite and the 8×8, 16×16, and 32×32 phased array antennas considered at each ground user. For each, shown is the azimuth cut when steered broadside, normalized by the maximum gain of each antenna. Shown also is the transmit beam mask for satellites operating at 20 GHz required by ITU [34].

We consider ground users equipped with various numbers of antennas, ranging between 8×8, 16×16, and 32×32 antenna arrays. In Fig. 2, we show the normalized azimuth cut of the gain delivered by the antenna array at each satellite and each of the ground user's receive antenna arrays under consideration. Included in Fig. 2 is the transmit beam mask, which is the maximum envelope of the antenna gain that satellites should abide by when transmitting, according to the International Telecommunication Union (ITU) [34]; our choice of antenna array at the satellite aligns quite well with this beam mask. We assume that satellites orient their antennas physically toward the center of the Earth and users orient their antennas upward toward an elevation of 90° (zenith) and then electronically beamform in their desired directions. All ground users are assumed to have a noise power spectral density of −174 dBm/Hz plus a 1.2 dB noise figure, based on recommendations published by 3rd Generation Partnership Project (3GPP) for rooftop-mounted users [35].

In order to assess worst-case interference, we assume the ground users are located at the same point. We found that similar conclusions are drawn even when users are separated by 5 km due to the large footprint of a satellite's beam on the surface of the Earth, and, in fact, this is in line with [16], which concluded that ground users separated by less than half the cell size could be considered co-located. Through simulation, we examine interference in multiple cities across the globe, namely Vancouver, Madrid, Seoul, Cape Town, Austin, Rio de Janeiro, and Bangalore. These cities vary widely in latitude from 10° to 50°, which has direct impact on the number of overhead satellites.

C. Interference Protection Constraint

The interference inflicted by secondary satellites onto primary ground users is of particular importance from a regulatory perspective. The precise definition of *prohibitive interference* is an open question with ongoing discussions

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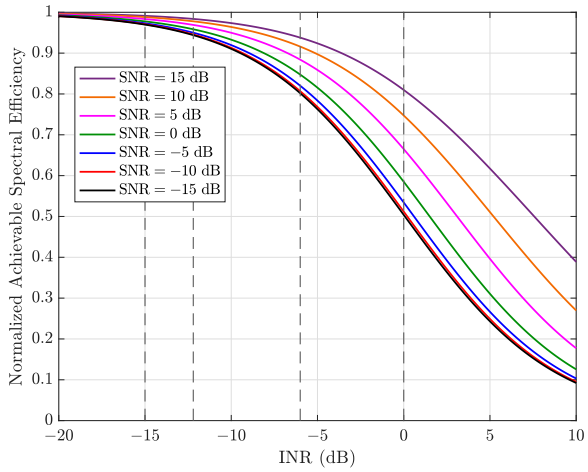

Fig. 3. Sacrifices in achievable spectral efficiency of a single link due to interference for various SNRs. Interference inflicted onto a primary system by the secondary system may only be tolerated if it is below a certain threshold; four thresholds considered in this work are shown as dashed lines.

and efforts to define such [11]. In this paper, we investigate one common approach based purely on the strength of interference [37], [38], [39]. As one motivating example, the ITU defines prohibitive interference as when the effective temperature of the receiver increases by more than 6% [10], [37] when treating interference as noise. In other words, the interference power at a primary user u must not be greater than 6% of its noise power $P_{\mathrm{n}}(\mathrm{u})$. Denoting $P_{\mathrm{int}}(\mathrm{u},\mathbf{p};\mathbf{s})$ as the interference power inflicted by a secondary satellite $\mathbf{s}$ onto a primary user u being served by $\mathbf{p}$, we can directly write this constraint as

$$\underbrace{\frac{P_{\mathrm{int}}(\mathrm{u},\mathbf{p};\mathbf{s})}{P_{\mathrm{n}}(\mathrm{u})}}_{=\ \mathsf{INR}(\mathrm{u},\mathbf{p};\mathbf{s})} \leq 0.06 \xrightarrow{\text{in dB}} \mathsf{INR}(\mathrm{u},\mathbf{p};\mathbf{s}) \leq \underbrace{-12.2\text{ dB}}_{\triangleq\ \mathsf{INR}_{\mathrm{th}}}, \tag{10}$$

where we use $\mathsf{INR}_{\mathrm{th}}$ to denote the interference threshold, in this case $-12.2\text{ dB} \approx 10\log_{10}(0.06)$. We refer to (10) as the *interference protection constraint*, which assume the secondary system is obligated to satisfy. Because there are active discussions as to what an acceptable level of interference is, this work will examine a few different values for $\mathsf{INR}_{\mathrm{th}}$, namely -15 dB, -12.2 dB, -6 dB, and 0 dB.

Beyond looking purely at interference, a constraint on throughput degradation has also been considered in the literature to more meaningfully quantify the impact of interference on a primary system [11]. This is naturally more difficult to quantify from the perspective of a secondary system, since it depends on the signal quality and modulation and coding schemes of the primary system, and thus introduces open questions on how to satisfy and evaluate such a protection constraint. In this work, we will consider a pure INR constraint for concreteness, but it should be noted that a throughput degradation constraint could be suitably mapped to an interference power constraint for a given primary signal quality.

Consider Fig. 3 illustrating the impact of INR on an arbitrary communication link. Shown is the degradation in achievable spectral efficiency $\log_2(1+\mathsf{SINR})$, normalized to the link's capacity $\log_2(1+\mathsf{SNR})$, as a function of INR for various SNR. When $\mathsf{INR} \leq -12.2$ dB, the link's achievable rate remains within 5% of the capacity for SNRs as low as -15 dB—arguably a justified tax for coexistence. However, the loss varies greatly with SNR and leads to a 50% reduction at low SNR when interference is as strong as noise at $\mathsf{INR} = 0$ dB.

IV. Bounds on the Interference Inflicted onto a Primary Ground User

A natural starting point to begin analyzing the feasibility of coexistence is to examine the upper and lower bounds on the amount of interference that secondary satellites may possibly inflict on a primary ground user. We investigate this by recording at each time step the minimum and maximum interference across all pairs of overhead primary and secondary satellites. The distribution of these bounds will then serve as a useful gauge on best- and worst-case interference scenarios. Actually attaining the lower bounds on interference would of course likely require cooperation or information exchange between the primary and secondary systems, but we are not necessarily suggesting such in this assessment.

Definition 1 (Absolute Bounds on INR): At a given instant, the maximum and minimum interference that can be inflicted onto a ground user u by the secondary system are respectively

$$\mathsf{INR}_{\max}(\mathrm{u}) = \max_{\mathbf{s}\in\mathcal{S}} \max_{\mathbf{p}\in\mathcal{P}} \mathsf{INR}(\mathrm{u},\mathbf{p};\mathbf{s}), \tag{11}$$

$$\mathsf{INR}_{\min}(\mathrm{u}) = \min_{\mathbf{s}\in\mathcal{S}} \min_{\mathbf{p}\in\mathcal{P}} \mathsf{INR}(\mathrm{u},\mathbf{p};\mathbf{s}). \tag{12}$$

These metrics shed light on the severity of interference when coexisting, and looking at their distributions over time will indicate how this severity varies as satellites in both systems progress along their orbits. The distribution of $\mathsf{INR}_{\max}(\mathrm{u})$, in particular, will dictate the need for an explicit interference protection constraint. For instance, if it is usually the case that $\mathsf{INR}_{\max}(\mathrm{u}) \leq -12.2$ dB, it may be argued that there is no need to consciously protect primary ground users; we will see shortly that this is not the case.

Definition 2 (Primary Satellite Selection): While the aforementioned *absolute* bounds are insightful, perhaps more useful are the maximum and minimum interference after the primary system has selected a satellite $\mathbf{p}^\star$ to serve its user u. Considering it is the burden of the secondary system to protect the primary system, we assume that the primary system performs satellite selection based purely on maximizing its own SNR. Put simply, $\mathbf{p}^\star$ is defined henceforth as

$$\mathbf{p}^\star = \arg\max_{\mathbf{p}\in\mathcal{P}} \mathsf{SNR}(\mathrm{u};\mathbf{p}). \tag{13}$$

In a real system, satellite selection would certainly involve a multitude of factors beyond maximizing SNR, but the exact algorithms employed by commercial systems remain proprietary and undisclosed to the public. With that being said, we observed little dependence on the particular criterion used to select the primary serving satellite.

Definition 3 (Conditional Bounds on INR): Now, with the assumption that a primary satellite selection $\mathbf{p}^\star$ has been made to serve u, the bounds on interference conditioned on this

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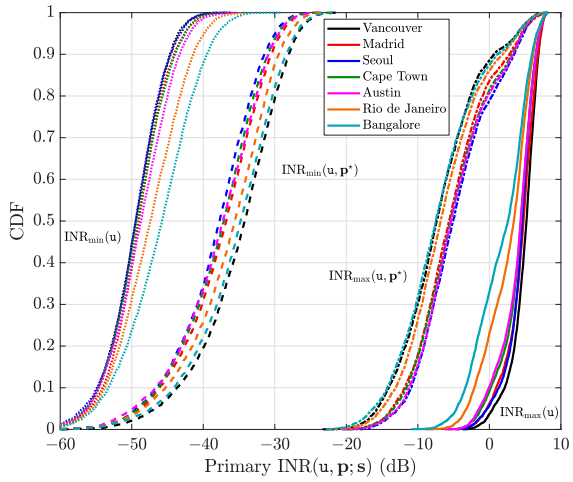

Fig. 4. The empirical CDFs of the absolute and conditional bounds on interference inflicted onto a primary ground user by the secondary system, evaluated at cities across the globe which vary widely in latitude.

TABLE IV
AVERAGE NUMBER OF OVERHEAD SATELLITES ACROSS THE GLOBE

City	Latitude	No. Primary Sat.	No. Secondary Sat.
Vancouver	49.2827°	28.29	10.35
Madrid	40.4168°	15.37	16.55
Seoul	37.5519°	13.95	18.76
Cape Town	−33.9249°	12.66	17.72
Austin	30.2672°	11.72	17.39
Rio de Janeiro	−22.9068°	10.45	12.98
Bangalore	12.9716°	9.52	10.81

selection are

$$\mathsf{INR}_{\max}(\mathrm{u}, \mathbf{p}^\star) = \max_{\mathbf{s}\in\mathcal{S}} \ \mathsf{INR}(\mathrm{u}, \mathbf{p}^\star; \mathbf{s}), \tag{14}$$

$$\mathsf{INR}_{\min}(\mathrm{u}, \mathbf{p}^\star) = \min_{\mathbf{s}\in\mathcal{S}} \ \mathsf{INR}(\mathrm{u}, \mathbf{p}^\star; \mathbf{s}). \tag{15}$$

From the perspectives of both systems, these are presumably more practical bounds since they take into account the important fact, as highlighted in (4), that interference at the primary user depends on both the secondary satellite $\mathbf{s}$ and its primary serving satellite $\mathbf{p}^\star$. It naturally follows that $\mathsf{INR}_{\max}(\mathrm{u}, \mathbf{p}^\star) \leq \mathsf{INR}_{\max}(\mathrm{u})$ and $\mathsf{INR}_{\min}(\mathrm{u}, \mathbf{p}^\star) \geq \mathsf{INR}_{\min}(\mathrm{u})$.

In Fig. 4, we plot the empirical CDFs of the absolute bounds $\mathsf{INR}_{\max}(\mathrm{u})$ and $\mathsf{INR}_{\min}(\mathrm{u})$ and the conditional bounds $\mathsf{INR}_{\max}(\mathrm{u}, \mathbf{p}^\star)$ and $\mathsf{INR}_{\min}(\mathrm{u}, \mathbf{p}^\star)$ across the globe for primary ground users equipped with 32×32 antenna arrays. Each CDF is taken over time, and the color of each denotes the users' location. Let us first examine the absolute bounds. The maximum interference $\mathsf{INR}_{\max}(\mathrm{u})$ is either just below or above the noise floor the vast majority of the time, and we observe that cities at higher latitudes (further from the equator) tend to see higher upper bounds $\mathsf{INR}_{\max}(\mathrm{u})$. This is perhaps best explained by the fact that there simply tend to be more primary-secondary satellite pairs at higher latitudes, illustrated in Table IV and attributed to Starlink's higher concentration of satellites at latitudes around 50° [40]. The steep distributions of $\mathsf{INR}_{\max}(\mathrm{u})$ suggest that—at virtually any given time and any location—there is a primary satellite $\mathbf{p}$ and a secondary satellite $\mathbf{s}$ in roughly the same direction as one another from the perspective of the primary ground user u. The minimum interference possible $\mathsf{INR}_{\min}(\mathrm{u})$ is extremely low across all cities, reliably at least 30 dB below the noise floor. In other words, there is virtually always some pair $(\mathbf{p} \in \mathcal{P}, \mathbf{s} \in \mathcal{S})$ that can guarantee low interference at a primary user u by virtue of low receive gain $G_{\mathrm{rx}}(\mathrm{u}, \mathbf{s}; \mathbf{p}) \approx 0$. Like with $\mathsf{INR}_{\max}(\mathrm{u})$, the minimum interference levels $\mathsf{INR}_{\min}(\mathrm{u})$ are also more extreme for higher-latitude users since there are more possible primary-secondary satellite pairs.

Takeaway 1: There is the Potential for Extremely High or Low Interference, Depending on the Serving Satellites: These results on the absolute bounds on interference in Fig. 4 suggest that there always exists the potential for extremely high interference if certain primary and secondary satellites are serving ground users in the same vicinity. On the flip side, however, there also always exists the potential for extremely low interference if a certain different pair of satellites is used to serve those same users.

Continuing an examination of Fig. 4, we now look at the bounds on interference *after* the primary serving satellite $\mathbf{p}^\star$ has been selected. The conditional minimum interference $\mathsf{INR}_{\min}(\mathrm{u}, \mathbf{p}^\star)$ is very low, more than 20 dB below the noise floor across cities at all times; this suggests that there is always a secondary satellite $\mathbf{s} \in \mathcal{S}$ that can offer low interference, even after the primary serving satellite $\mathbf{p}^\star$ has been chosen. Likewise, however, based on the conditional maximum $\mathsf{INR}_{\max}(\mathrm{u}, \mathbf{p}^\star)$, there is always the potential for a secondary satellite to inflict substantial interference that exceeds a threshold of −12.2 dB, for example, laid forth before based on ITU recommendations [10].

Having conditioned on the primary serving satellite, the variability across cities is largely dictated by the number of secondary satellites overhead, whose average is listed in Table IV obtained by simulation. Cities with more secondary satellites overhead tend to see higher $\mathsf{INR}_{\max}(\mathrm{u}, \mathbf{p}^\star)$ and lower $\mathsf{INR}_{\min}(\mathrm{u}, \mathbf{p}^\star)$. In this particular case, Kuiper's constellation has fewer satellites at higher absolute latitudes and near the equator, so these satellites see less extreme $\mathsf{INR}_{\max}(\mathrm{u}, \mathbf{p}^\star)$ and $\mathsf{INR}_{\min}(\mathrm{u}, \mathbf{p}^\star)$. In general, the distributions for both $\mathsf{INR}_{\max}(\mathrm{u}, \mathbf{p}^\star)$ and $\mathsf{INR}_{\min}(\mathrm{u}, \mathbf{p}^\star)$ across cities are fairly similar in shape and both differ in median by a few decibels. For the sake of concreteness, our results henceforth will be based on users located in Austin, Texas, with a latitude-longitude of (30.267153°, −97.743057°), which is where we see worst-case $\mathsf{INR}_{\max}(\mathrm{u}, \mathbf{p}^\star)$, along with Seoul.

Takeaway 2: Strategically Selecting a Secondary Satellite Can Reliably Protect Primary Ground Users: Even if the primary system performs satellite selection completely independently of the secondary system, the distribution of $\mathsf{INR}_{\min}(\mathrm{u}, \mathbf{p}^\star)$ suggests that it is still viable for the secondary to strategically select its serving satellite such that low interference is inflicted on the primary ground user. Its ability to do so in a manner that delivers sufficiently high downlink SINR to its own ground user is an open question we examine further in the sections that follow.

V. SYSTEM PERFORMANCE UNDER A STRICT INTERFERENCE PROTECTION CONSTRAINT

Beyond solely examining interference, this section aims to gauge both primary and secondary system performance

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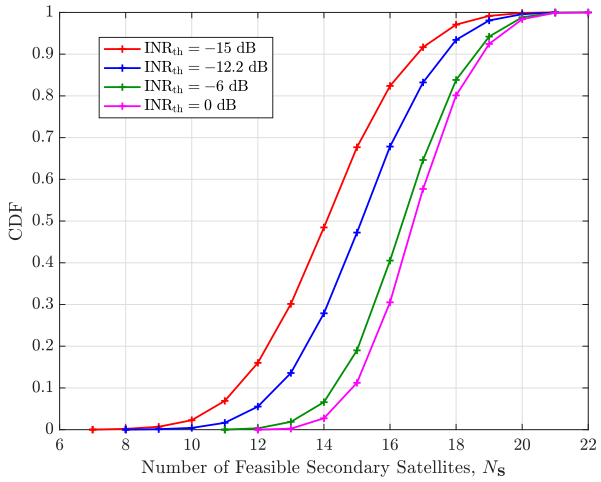

Fig. 5. The empirical CDF (over time) of the number of feasible secondary satellites $N_{\mathbf{s}}$ satisfying the interference protection constraint for various thresholds $\mathsf{INR}_{\mathrm{th}}$.

in terms of SINR. To accomplish this, we consider various criteria by which the secondary system selects its serving satellite $\mathbf{s} \in \mathcal{S}$ to serve a ground user v. As with this entire paper, our goal here is not to propose a technique maximizing system performance but rather to evaluate the merits of particular techniques by investigating the interference and downlink signal quality they attain. This will serve as a useful first step in assessing the feasibility of coexistence, with the goal of deriving insights that may motivate more sophisticated techniques in future work.

Definition 4 (Interference Protection Constraint): When the secondary system selects its serving satellite $\mathbf{s} \in \mathcal{S}$ in the presence of a primary user u served by a primary satellite $\mathbf{p}^\star$, imposing an interference protection constraint amounts to this selection satisfying

$$\mathsf{INR}(\mathrm{u}, \mathbf{p}^\star; \mathbf{s}) \leq \mathsf{INR}_{\mathrm{th}}. \tag{16}$$

This is a strict instantaneous constraint but could take other forms, potentially involving how frequently the threshold is violated—e.g., across time, frequency, and or space—making for interesting future investigation.

Definition 5 (Number of Feasible Secondary Satellites): A natural first question we investigate is the following: How many secondary satellites overhead are capable of satisfying the protection constraint? This can be formally expressed as the number of feasible secondary satellites $N_{\mathbf{s}}$ defined as

$$N_{\mathbf{s}} = |\{\mathbf{s} \in \mathcal{S} : \mathsf{INR}(\mathrm{u}, \mathbf{p}^\star; \mathbf{s}) \leq \mathsf{INR}_{\mathrm{th}}\}|, \tag{17}$$

where $|\mathcal{A}|$ denotes the cardinality of a set $\mathcal{A}$.

Recording $N_{\mathbf{s}}$ throughout the duration of our simulation as both the primary and secondary satellites orbit allows us to populate its empirical CDF in Fig. 5 for various thresholds $\mathsf{INR}_{\mathrm{th}}$. It can be seen that over half the time there are at least 15 secondary satellites capable of inflicting interference less than -12.2 dB when serving their ground user v. In other words, about 55% of the time, the secondary system has its choice between 15 or more satellites when forced to meet a strict interference constraint of -12.2 dB. This number of feasible satellites reduces with a stricter threshold, whereas relaxing the threshold to -6 dB, for instance, adds roughly two satellites to the pool in median. In all cases, there are rare but potentially concerning instances where only 6–12 satellites can meet the threshold, imposing limited flexibility in the secondary system's ability to serve its ground user.

Definition 6 (Greedy Max-SNR and Max-SINR Selection): Let us begin by considering two greedy secondary satellite selection approaches that ignore the interference protection constraint. We first consider the case where the secondary system purely maximizes its own SNR by making its satellite selection as

$$\mathbf{s}^\dagger_\infty = \arg\max_{\mathbf{s}\in\mathcal{S}} \ \mathsf{SNR}(\mathrm{v}, \mathbf{s}). \tag{18}$$

Here, the use of ∞ can be thought of as representing an infinite interference threshold $\mathsf{INR}_{\mathrm{th}}$. Similarly, we also consider the case where the secondary system maximizes its own SINR as

$$\mathbf{s}^\star_\infty = \arg\max_{\mathbf{s}\in\mathcal{S}} \ \mathsf{SINR}(\mathrm{v}, \mathbf{s}; \mathbf{p}^\star). \tag{19}$$

It is valuable to consider both of these selections because, given the number of feasible secondary satellites $N_{\mathbf{s}}$, it may be the case that the secondary system *inherently* satisfies the interference protection constraint when purely maximizing its own SNR or SINR. We will see this is not the case, however.

In Fig. 6, we evaluate the performance of both the primary and secondary systems under the greedy max-SNR and greedy max-SINR secondary satellite selection schemes. First, let us consider Fig. 6a, showing the empirical CDFs of primary SINR over time under each secondary selection scheme for various receive antenna arrays.[6] The upper bounds on SINR (i.e., the interference-free SNR) for each array size is shown as the dotted line. For all three array sizes, purely maximizing SINR at the secondary system tends to yield primary system performance superior than purely maximizing SNR. Somewhat intuitively, it can thus be concluded that improving the secondary system's SINR results in the primary system also enjoying higher SINR; in other words, by choosing a secondary satellite that reduces the interference inflicted onto its own ground user v by the selected primary satellite $\mathbf{p}^\star$, the interference inflicted onto the primary user u is also inherently reduced. The superiority of max-SINR over max-SNR is most clearly seen in lower tail improvement, which has historically been a point of concern for network service providers.

Now, we consider Fig. 6b, depicting secondary system SINR analogous to Fig. 6a. Many of the same trends can be observed but are more extreme. The improvement offered by maximizing SINR is more obvious as it drastically reduces the density of the lower tail and pulls the distribution quite close to its upper bound. Clearly, maximizing its own SINR would yield the preferred outcome from the perspective of the secondary system but this does not guarantee sufficiently low interference at the primary user at all times. This can be seen in Fig. 7, which compares the primary INR under both secondary selection schemes. While maximizing the secondary

[6]For concreteness, we use SINR as a representative system performance metric. Other metrics, such as throughput, can be derived from the SINR considering other system factors such as modulation and coding.

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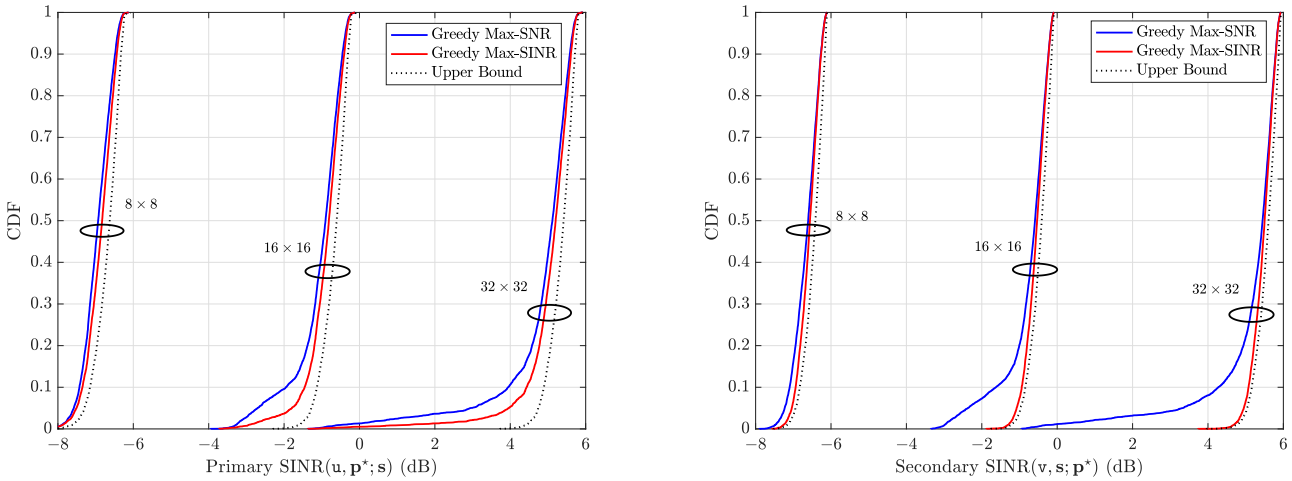

(a) Primary ground user SINR. (b) Secondary ground user SINR.

Fig. 6. The CDFs of (a) primary SINR and (b) secondary SINR over time under greedy max-SNR and greedy max-SINR secondary satellite selection $\mathbf{s}^{\dagger}_{\infty}$ and $\mathbf{s}^{\star}_{\infty}$ for various ground user antenna array sizes. For each, the upper bound of SINR is shown as a dotted line, corresponding to the interference-free SNR. Choosing a secondary satellite to maximize its SINR inherently also improves the primary SINR.

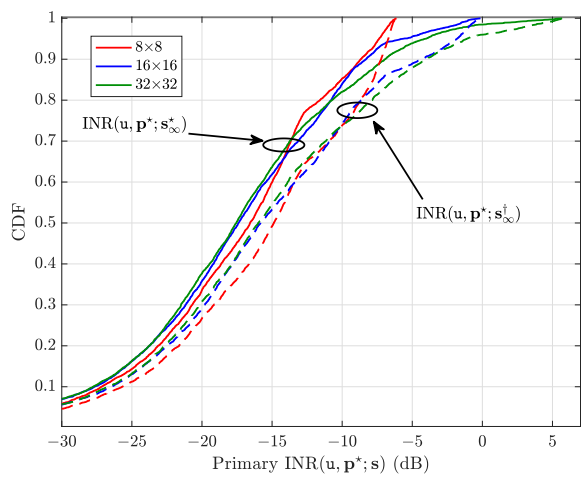

Fig. 7. The interference inflicted onto a primary user served by a satellite $\mathbf{p}^{\star}$ by a secondary satellite which purely maximizes its own SNR or maximizes its own SINR, without protecting the primary user.

SINR results in lower primary interference than maximizing SNR, a signification portion of the time, interference is well above the thresholds under consideration (e.g., -12.2 dB). In Fig. 7, we also observe a general increase in the upper tail of interference as the array size increases, since more antennas leads to a higher receive gain in instances when the primary and secondary serving satellites are in the same vicinity.

Takeaway 3: Max-SINR is Preferable for Both Primary and Secondary Systems, But Does Not Guarantee Protection: The secondary system selecting the satellite which maximizes its SINR is preferred over maximizing its SNR from the perspective of both the secondary system and the primary system, since it will inherently reduce interference at both systems' ground users. Doing so, however, does not guarantee that the interference inflicted onto the primary system will be below a plausible protection threshold. This motivates the need to explicitly protect the primary ground user when selecting the secondary serving satellite.

Definition 7 (Protective Max-SNR and Max-SINR Selection): We now augment the previous selection methodologies by enforcing the interference protection constraint (16) to form *protective* max-SNR selection

$$\mathbf{s}^{\dagger} = \arg\max_{\mathbf{s}\in\mathcal{S}} \ \mathsf{SNR}(\mathbf{v},\mathbf{s}) \tag{20a}$$

$$\text{s.t.} \ \ \mathsf{INR}(\mathbf{u},\mathbf{p}^{\star};\mathbf{s}) \leq \mathsf{INR}_{\text{th}} \tag{20b}$$

and *protective* max-SINR selection

$$\mathbf{s}^{\star} = \arg\max_{\mathbf{s}\in\mathcal{S}} \ \mathsf{SINR}(\mathbf{v},\mathbf{s};\mathbf{p}^{\star}) \tag{21a}$$

$$\text{s.t.} \ \ \mathsf{INR}(\mathbf{u},\mathbf{p}^{\star};\mathbf{s}) \leq \mathsf{INR}_{\text{th}}. \tag{21b}$$

Outright maximizing SINR via (21) would presumably yield superior secondary system performance over (20), but it very well may be the case that meeting the interference protection constraint and maximizing its own SNR would inherently improve its own SINR. This makes it interesting to compare these selections and their resulting performance.

In Fig. 8, we compare the performance of both of these protective secondary satellite selection schemes for various interference thresholds INR_{th} with 32×32 antenna arrays at the ground users. First and foremost, comparing Fig. 8a to Fig. 6a illustrates that incorporating the protective constraint improves primary SINR, greatly reducing the lower tail and highlighting the importance of explicitly incorporating the protection constraint into secondary satellite selection. Aside from this, protective max-SINR selection clearly improves primary SINR over protective max-SNR, magnifying when INR_{th} is relaxed. This is again due to the inherent reductions in interference at the primary user when maximizing the secondary SINR; in this case, sometimes those reductions push interference further below INR_{th}, but this is typically only observed when INR_{th} is less strict.

Now, in Fig. 8b, we derive important conclusions on secondary system performance when abiding by the protection constraint. The shown upper bound is the interference-free SNR. The first key observation is that both protective

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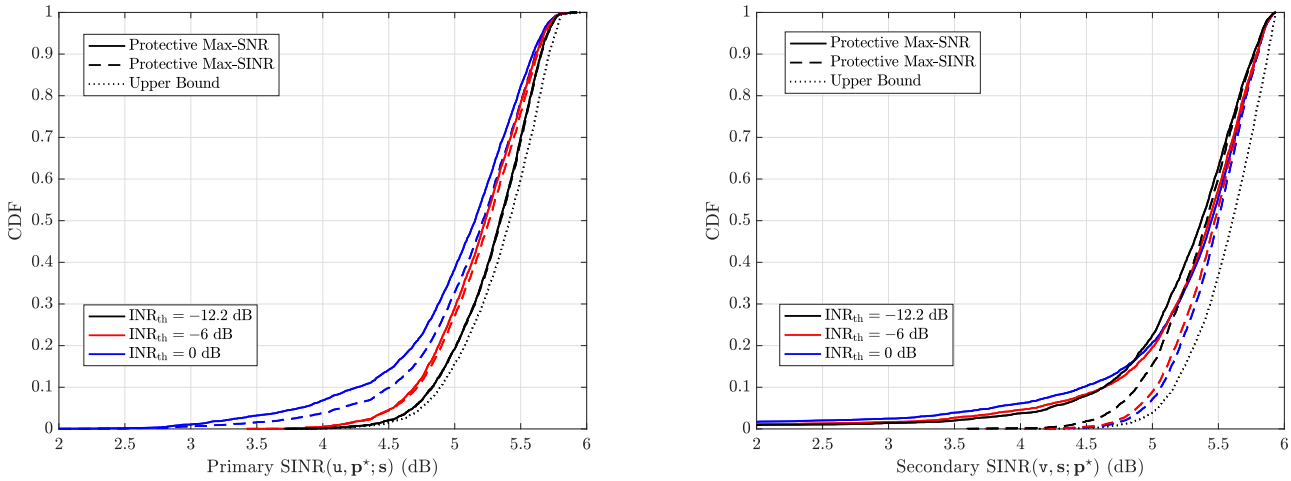

(a) Primary ground user SINR. (b) Secondary ground user SINR.

Fig. 8. The CDF of SINR of primary and secondary users with 32×32 phased array antennas per INR threshold where INR threshold is color-coded as black: $\mathsf{INR}_{\text{th}} = -12.2$ dB, red: $\mathsf{INR}_{\text{th}} = -6$ dB, and blue: $\mathsf{INR}_{\text{th}} = 0$ dB. While protecting a primary user, choosing the secondary satellite which maximizes SINR offers far superior secondary and slightly better primary performance than maximizing SNR.

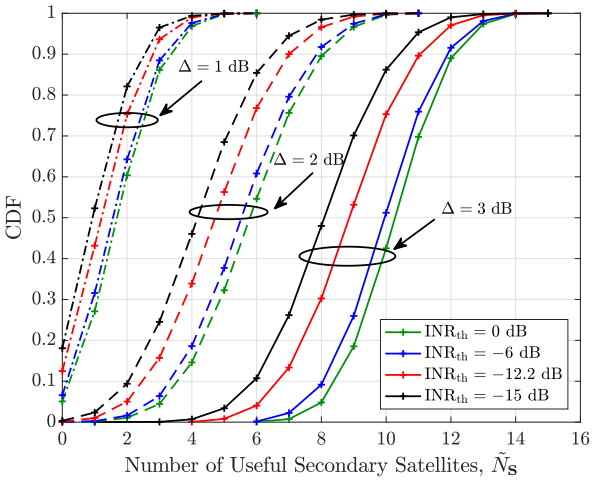

Fig. 9. The empirical CDF (over time) of the number of useful secondary satellites $\tilde{N}_{\mathbf{s}}$ satisfying the protection constraint while offering a secondary SINR within a factor of Δ of the maximum SNR.

max-SNR and protective max-SINR are capable of delivering appreciable SINRs the overwhelming majority of the time. Protective max-SNR exhibits a heavier tail, since the only interference reduction it enjoys is the inherent reduction when satisfying the protection constraint. Protective max-SINR, however, impressively sees minor degradation even for the most stringent constraint. As the constraint is relaxed, protective max-SINR approaches the upper bound but ultimately falls short due to interference it incurs from the primary system.

Takeaway 4: Protective Max-SINR is Preferable for Both Systems and Guarantees Primary Users are Protected: Protecting a primary ground user does not necessarily impede the secondary system from attaining appreciable SINR, falling short of its upper bound by fractions of a decibel. The gains in secondary system performance by maximizing its SINR while meeting the protection constraint are non-negligible compared to maximizing its SNR. As an added benefit, when the secondary system maximizes its SINR, the primary system also enjoys an increase in SINR.

It is certainly a welcome sight that impressive performance can be achieved while meeting very stringent protection constraints, but it is perhaps more useful to understand how many secondary satellites are capable of such, since systems will naturally be tasked with serving multiple users with multiple satellites. We investigate this further in the following.

Definition 8 (Number of Useful Secondary Satellites): Definition 5 introduced $N_{\mathbf{s}}$, the number of secondary satellites capable of satisfying the protection constraint. We extend this notion to define the number of *useful* secondary satellites $\tilde{N}_{\mathbf{s}}$, capable of satisfying the protection constraint while also offering a secondary SINR within a factor of Δ from the maximum possible SNR (i.e., $\mathsf{SNR}(\mathrm{v}, \mathbf{s}^{\dagger}_{\infty})$),

$$\tilde{N}_{\mathbf{s}} = \big|\big\{\mathbf{s} \in \mathcal{S} : \mathsf{INR}(\mathrm{u}, \mathbf{p}^{\star}; \mathbf{s}) \leq \mathsf{INR}_{\text{th}}, \\ \mathsf{SINR}(\mathrm{v}, \mathbf{s}; \mathbf{p}^{\star}) \geq \mathsf{SNR}(\mathrm{v}, \mathbf{s}^{\dagger}_{\infty}) \cdot \Delta^{-1}\big\}\big|, \tag{22}$$

Fig. 9 exhibits the empirical CDF of the number of useful secondary satellites for various Δ and various thresholds INR_{th}, where Δ can be interpreted as the maximum amount of SINR the secondary system is willing to sacrifice from the upper bound SINR, i.e., the interference-free SNR, to protect the primary ground user. Earlier in Fig. 5, we saw that there were typically 12–18 feasible secondary satellites, depending on the protection constraint. Comparing this to Fig. 9, we can see that this decreases to 8–12 satellites if we wish to lose at most $\Delta = 3$ dB in secondary SINR. This decreases further to 4–8 satellites with $\Delta = 2$ dB and to 1–3 satellites with $\Delta = 1$ dB.

Takeaway 5: Multiple Satellites are Useful for Coexistence at any Given Time: While there are often 12–18 secondary satellites which can satisfy even the most stringent protection constraint, typically 1–3 of these satellites can also deliver near-maximal SINR. This number grows to 4–12 satellites, however, if 2–3 dB in SINR loss can be tolerated.

Definition 9 (Satellite Angular Separation): In order to derive real-world meaning from these four different techniques

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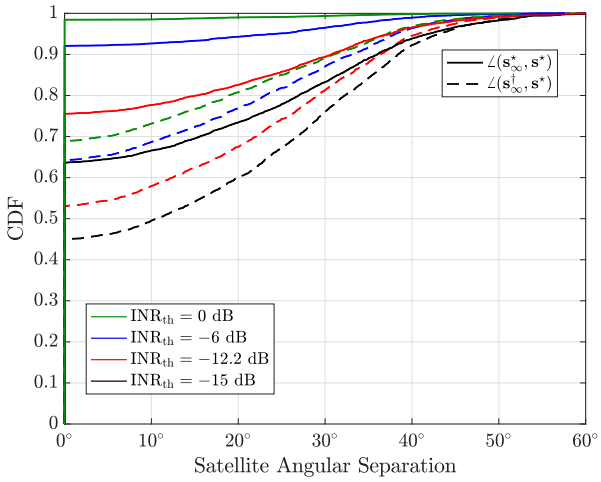

Fig. 10. For various $\mathsf{INR}_{\mathrm{th}}$, shown are the empirical CDFs (over time) of the angular separation between the protective max-SINR secondary satellite selection $\mathbf{s}^\star$ and (i) the greedy max-SINR selection, $\angle(\mathbf{s}^\star_\infty, \mathbf{s}^\star)$; and (ii) the greedy max-SNR, $\angle(\mathbf{s}^\dagger_\infty, \mathbf{s}^\star)$.

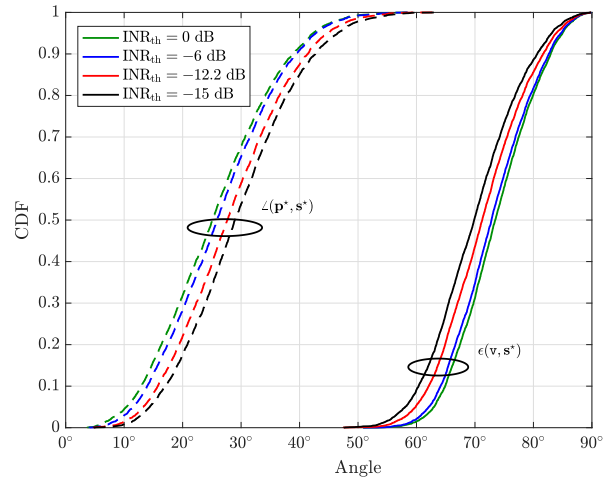

Fig. 11. The solid lines depict the empirical CDFs (over time) of the elevation angle of the secondary satellite $\mathbf{s}^\star$ which maximizes its SINR while protecting the primary ground user. The dashed lines correspond to that of the angular separation between $\mathbf{s}^\star$ and the primary serving satellite $\mathbf{p}^\star$.

for secondary satellite selection, we introduce the notion of angular separation between two satellites. Let us define $\angle(\mathbf{s}_1, \mathbf{s}_2)$ as the absolute angular separation between two satellites $\mathbf{s}_1$ and $\mathbf{s}_2$ with respect to the primary ground user $\mathbf{u}$.

In Fig. 10, the solid lines depict the distributions (over time) of the angular separation between the protective max-SINR selection $\mathbf{s}^\star$ and the greedy max-SINR selection $\mathbf{s}^\star_\infty$ for various $\mathsf{INR}_{\mathrm{th}}$. The dashed lines correspond to the angular separation between the protective max-SINR selection $\mathbf{s}^\star$ and the greedy max-SNR selection $\mathbf{s}^\dagger_\infty$. From the solid green line, we can conclude that about 99% of the time $\mathbf{s}^\star = \mathbf{s}^\star_\infty$; in other words, the protection constraint of $\mathsf{INR}_{\mathrm{th}} = 0$ dB is met inherently by simply maximizing the secondary system SINR. As the protection constraint is tightened to $\mathsf{INR}_{\mathrm{th}} = -12.2$ dB (red solid line), only about 75% of time does $\mathbf{s}^\star = \mathbf{s}^\star_\infty$. The remaining roughly 25% of the time, the two will typically be separated by 10° to 40°. Comparing the solid green line to the dashed green line, we can see that it is far less common yet still frequent that maximizing the secondary system SNR also inherently satisfies the protection constraint; as a result, there tends to be more angular separation.

Now, in Fig. 11, the dotted lines show the angular separation between the primary serving satellite $\mathbf{p}^\star$ and the protective max-SINR secondary satellite $\mathbf{s}^\star$. In general, the two satellites are separated by 15° to 45°, with the separation tending to increase as the protection constraint is made more stringent. The solid lines depict the elevation angle of the secondary satellite $\mathbf{s}^\star$. With a relaxed protection constraint, the secondary SINR is typically maximized by selecting a satellite at higher elevations, typically 65° to 85°. As the protection constraint is made more stringent, the SINR-maximizing satellite is typically at elevations about 5° lower, closer to the horizon.

Takeaway 6: As the Constraint Becomes More Stringent, the Secondary System Selects a Satellite With a Larger Angular Separation from the Primary Serving Satellite to Meet the Required Protection Level: While greedy max-SINR approach inherently satisfies the protection constraint more often than greedy max-SNR, it still fails to meet the constraint 25% of the time when the threshold is tightened to $\mathsf{INR}_{\mathrm{th}} = -12.2$ dB. To comply with the protection constraint, greedy max-SINR tends to choose a serving satellite that is farther from both greedy max-SNR selection and the primary serving satellite. As the protection constraint becomes stricter, the elevation angle of the selected satellite decreases.

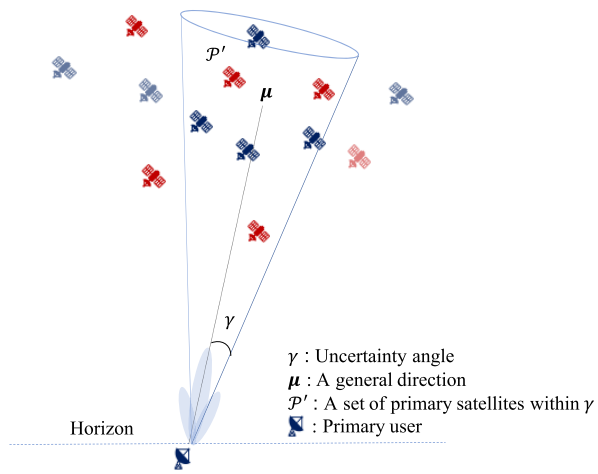

Fig. 12. Uncertainty in primary satellite selection. Any satellites within an angular distance of γ from the direction $\boldsymbol{\mu}$ are considered potential serving satellites for the primary user.

VI. System Performance Under Uncertainty

In the previous section, we characterized system performance under secondary satellite selection techniques which protect a primary user $\mathbf{u}$ receiving downlink from its serving satellite $\mathbf{p}^\star$. Among others, one noteworthy practical challenge in these secondary satellite selection problems is the assumption that the secondary system has knowledge of $\mathbf{p}^\star$. This section explores when this is not the case and how such uncertainty may impact system performance.

Definition 10 (Uncertainty in Primary Satellite Selection): Suppose rather than knowing for certain that the primary

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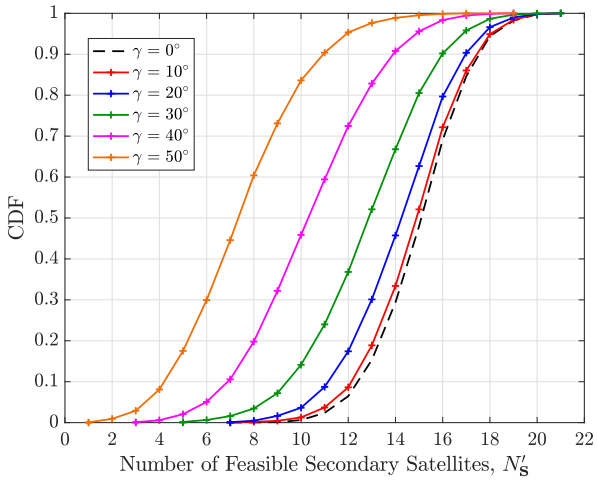
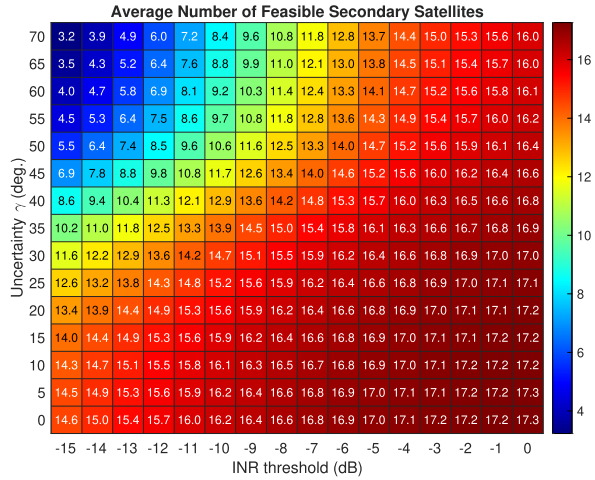


(a) Number of feasible satellites under uncertainty.



(b) Number of feasible secondary satellites under uncertainty.

Fig. 13. Shown in (a) are the empirical CDFs of the number of feasible secondary satellites $N'_{\mathbf{s}}$ capable of satisfying a protection constraint of $\mathsf{INR}_{\text{th}} = -12.2$ dB for all $\mathbf{p}^\star \in \mathcal{P}'$ under various levels of uncertainty γ. In (b), the average number of feasible secondary satellites is shown for various levels of uncertainty γ and protection constraints INR_{th}. Even in the most extreme case, there are more than 3 feasible secondary satellites on average.

ground user u is being served by $\mathbf{p}^\star$, the secondary system instead only knows that the true serving satellite $\mathbf{p}^\star$ is within some set of primary satellites $\mathcal{P}' \subseteq \mathcal{P}$. To model this uncertainty, suppose the secondary system knows (or is confident) that the angular separation of the true primary serving satellite $\mathbf{p}^\star \in \mathcal{P}$ and some direction $\boldsymbol{\mu}$ is within γ. In other words, any primary satellites located within an angular distance of γ from the direction $\boldsymbol{\mu}$ are considered potential serving satellites for the primary user. Then, the set of possible serving satellites $\mathcal{P}'$ can be expressed as follows and is illustrated in Fig. 12.

$$\mathcal{P}' = \{\mathbf{p} \in \mathcal{P} : \angle(\boldsymbol{\mu}, \mathbf{p}) \leq \gamma\} \tag{23}$$

As before, the absolute angular difference operation $\angle(\cdot)$ is from the perspective of the primary ground user u.

It is worth emphasizing that, at any given time, it is reasonable to assume that the secondary system has near-real-time knowledge of all primary satellite locations $\mathcal{P}$ since this is public information; this could perhaps be used by the secondary system to populate $\mathcal{P}'$, assuming it can acquire some estimate $\boldsymbol{\mu}$ on the vicinity of the true primary serving satellite. For the sake of simulation, we construct this set $\mathcal{P}'$ at a given instant by considering all satellites within some angular distance γ of the primary satellite which maximizes SNR, defined before as $\mathbf{p}^\star$ in (13); in other words, we take $\boldsymbol{\mu} = \mathbf{p}^\star$. Note that we have not assumed the true primary serving satellite to necessarily be in the direction of $\boldsymbol{\mu}$ nor will it be relevant in the results that follow, since we will focus on *worst-case* secondary system performance under uncertainty.

Definition 11 (Protection Constraint under Uncertainty): When faced with complete uncertainty about which primary satellite within the set $\mathcal{P}'$ is serving user u, ensuring a selected secondary satellite $\mathbf{s}$ does not exceed a threshold INR_{th} amounts to the constraint

$$\mathsf{INR}(\mathrm{u}, \mathbf{p}; \mathbf{s}) \leq \mathsf{INR}_{\text{th}} \ \forall \ \mathbf{p} \in \mathcal{P}'. \tag{24}$$

Naturally, this is a stricter protection constraint than (16), when the secondary system has knowledge of $\mathbf{p}^\star$, since it must not inflict prohibitively high interference for several potential primary serving satellites.

Definition 12: Number of Feasible Satellites Under Uncertainty: Under uncertainty, the number of secondary satellites satisfying this stricter protection constraint is

$$N'_{\mathbf{s}} = |\{\mathbf{s} \in \mathcal{S} : \mathsf{INR}(\mathrm{u}, \mathbf{p}; \mathbf{s}) \leq \mathsf{INR}_{\text{th}} \ \forall \ \mathbf{p} \in \mathcal{P}'\}|. \tag{25}$$

Naturally, we have $N'_{\mathbf{s}} \leq N_{\mathbf{s}}$ as a consequence of uncertainty. In Fig. 13, we highlight how the level of uncertainty γ and the interference threshold INR_{th} impact the number of feasible secondary satellites $N'_{\mathbf{s}}$. First, in Fig. 13a, we depict the empirical CDF of $N'_{\mathbf{s}}$ over time for various γ when the threshold is $\mathsf{INR}_{\text{th}} = -12.2$ dB. Notice that the shape of the distribution remains fairly unchanged but undergoes a shift leftward as uncertainty γ increases. In median, there are typically more than 4 fewer feasible satellites when $\gamma = 40°$, compared to when $\mathbf{p}^\star$ is known exactly, and 8 fewer when $\gamma = 50°$. In Fig. 13b, we depict a heatmap of the time-averaged $N'_{\mathbf{s}}$ for a wide range of γ and INR_{th}. Even with extreme uncertainty and a very strict threshold, on average there are 3 satellites that can guarantee the protection constraint is met across the entire set of possible primary satellites $\mathcal{P}'$. At $\mathsf{INR}_{\text{th}} = -12$ dB, there are more than 10 feasible satellites on average when uncertainty is $40°$ or less. With at most $25°$ of uncertainty, the average number of feasible satellites decreases from 17 to 12, compared to a virtually unconstrained and perfectly known scenario (i.e., when $\mathsf{INR}_{\text{th}} = 0$ dB and $\gamma = 0°$).

Even though we only assume one secondary satellite is chosen to serve a ground user v, it is useful to examine how many satellites are feasible, since it sheds light on the flexibility the secondary system has in meeting the constraint. For instance, considering a practical secondary system would be tasked with serving multiple users at once across its constellation of satellites, it may benefit greatly from having more satellites capable of meeting a strict protective constraint at any given time. If $N'_{\mathbf{s}} = 1$, for example, the secondary system would be forced to serve the user with the lone feasible

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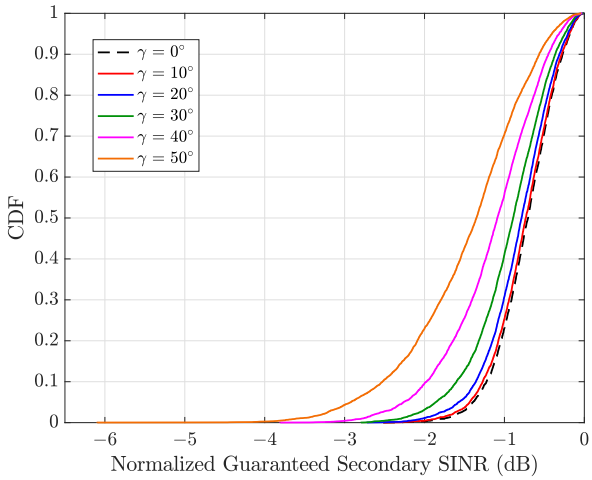
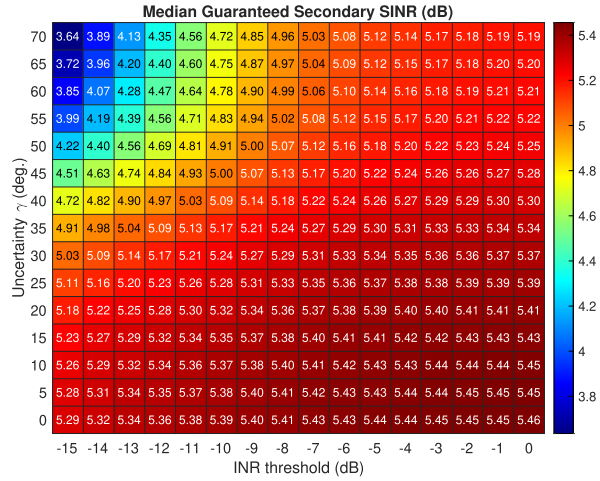


(a) Guaranteed secondary SINR under uncertainty.



(b) Median guaranteed secondary SINR under uncertainty.

Fig. 14. Shown in (a) are the empirical CDFs (over time) of the guaranteed SINR of the secondary system, normalized to its upper bound (SNR), under various levels of uncertainty γ when $\mathsf{INR}_{\text{th}} = -12.2$ dB. In (b), the median guaranteed SINR of the secondary system is shown for various levels of uncertainty and protection constraints INR_{th}. There typically exists secondary satellites which guarantee appreciable SINR while protecting primary users.

satellite or not serve the user at all if the satellite is occupied serving another user. This also highlights how multi-beam satellites may relax the challenges associated with this process.

Takeaway 7: Multiple Secondary Satellites are Feasible Under a Stringent Protection Constraint and Uncertainty: Even with a very stringent interference protection constraint such as -12.2 dB, there are on average more than 10 secondary satellites capable of protecting a primary ground user under modest knowledge of its serving satellite. Likewise, with extreme uncertainty about the primary serving satellite, there are more than 10 secondary satellites capable of protecting a primary ground user under a modest interference constraint.

Definition 13 (Protective Max-Guaranteed-SINR Selection): We now investigate the effects of this uncertainty on secondary system performance. To do so, we introduce the notion of maximum *guaranteed* secondary system SINR under uncertainty, which can be accomplished by solving the following constrained satellite selection problem.

$$(\mathbf{s}', \mathbf{p}') = \arg\max_{\mathbf{s}\in\mathcal{S}} \min_{\mathbf{p}\in\mathcal{P}'} \mathsf{SINR}(\mathrm{v}, \mathbf{s}; \mathbf{p}) \tag{26a}$$

$$\text{s.t.} \quad \mathsf{INR}(\mathrm{v}, \mathbf{p}; \mathbf{s}) \leq \mathsf{INR}_{\text{th}} \ \forall\ \mathbf{p} \in \mathcal{P}' \tag{26b}$$

Notice that $\mathsf{SINR}(\mathrm{v}, \mathbf{s}'; \mathbf{p}')$ is the minimum SINR that the secondary system will see; we refer to this as the *guaranteed* SINR, since for any $\mathbf{p} \in \mathcal{P}'$

$$\mathsf{SINR}(\mathrm{v}, \mathbf{s}'; \mathbf{p}') \leq \mathsf{SINR}(\mathrm{v}, \mathbf{s}'; \mathbf{p}) \ \forall\ \mathbf{p} \in \mathcal{P}'. \tag{27}$$

In Fig. 14a, we show the distribution of guaranteed SINR achieved by the secondary system over time, normalized to its upper bound, the maximum SNR (i.e., $\mathsf{SNR}(\mathrm{v}, \mathbf{s}^{\dagger}_{\infty})$ from (18)), where $\mathsf{INR}_{\text{th}} = -12.2$ dB. By normalizing the guaranteed SINR in this way, we can measure the degradation in secondary system signal quality due to the combined effects of (i) abiding by the protection constraint under uncertainty and (ii) interference from the primary system. Without any uncertainty $\gamma = 0°$, there is at worst about a 2.5 dB gap in guaranteed SINR from its upper bound and less than a 1 dB gap the majority of the time. When $\mathbf{p}^{\star}$ is not known precisely, this uncertainty leads to losses in guaranteed SINR, especially at the lower tails, but this is most apparent only under high uncertainty. For $\gamma \leq 20°$, the guaranteed secondary SINR falls short by only a fraction of a dB from that with perfect knowledge of $\mathbf{p}^{\star}$. When uncertainty increases to $\gamma = 50°$, there is only about a 1 dB sacrifice in median, but a much more substantial lower tail is present. Nonetheless, it is a welcome sight that, the overwhelming majority of the time, less than 4 dB is sacrificed by the secondary system in protecting (and being interfered by) the primary system, even with very limited knowledge on the primary serving satellite.

Fig. 14b extends this analysis by depicting the median guaranteed secondary SINR(unnormalized) for various interference thresholds and levels of uncertainty. It can be observed in the most extreme case that the minimum guaranteed SINR is above 3.5 dB in median, about 2 dB short of the upper bound. In terms of median guaranteed SINR, the secondary system is not significantly impacted by making the protection threshold more strict under modest uncertainty. When uncertainty exceeds $40°$, however, we begin to see that more stringent INR_{th} leads to more dramatic losses in median guaranteed SINR.

Takeaway 8: With Limited Knowledge About the Primary Serving Satellite, the Secondary System Can Still Protect Primary Ground Users and Deliver High SINR: Even when the secondary system does not know precisely which primary satellite is serving a particular ground user, if it has limited knowledge on the general vicinity of the primary serving satellite, it can still select a satellite which guarantees the primary user is protected and delivers appreciable SINR. Although, uncertainty leads to wider variability in the SINR the secondary system is capable of guaranteeing, the severity of such is most apparent under fairly high uncertainty. With that being said, the number of feasible satellites does decrease, limiting the freedom a practical secondary system would have in scheduling its satellites when serving multiple ground users and protecting the primary system.

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From Fig. 14a, it is certainly remarkable at first glance that the secondary system sacrifices at most only 4 dB in guaranteed SINR under an uncertainty of $\gamma = 50^\circ$ and a strict interference constraint of $\mathsf{INR}_{\mathrm{th}} = -12.2$ dB, compared to exact knowledge of $\mathbf{p}^\star$. The SNR delivered by a secondary satellite $\mathbf{s} \in \mathcal{S}$ is dictated mostly by its path loss to the ground user, which itself depends on the satellite altitude and elevation angle [35]. Across all overhead satellites, there is typically 4–5 dB of variability in SNR delivered to a given user [41]. Combining this with the fact that satisfying the interference protection constraint often inherently reduces interference onto the secondary ground user, a *feasible* secondary satellite typically only sacrifices at most roughly 4–5 dB in SINR when $\mathsf{INR}_{\mathrm{th}}$ is strict. Selecting the feasible satellite which maximizes SINR therefore often results in less than 4–5 dB of loss.

Takeaway 9 (Cooperation can help close the gap): Even with these encouraging conclusions drawn, it should nonetheless be noted that fairly minimal cooperation between the primary and secondary systems could close the gap we observe under uncertainty. Sharing the steering directions of the satellites between systems could foster a healthy coexistence paradigm, allowing secondary systems to better protect the primary system's ground users while reliably serving its own.

VII. Conclusion and Future Direction

This work has investigated the feasibility of in-band coexistence between two heterogeneous LEO satellite communication systems by analyzing two preeminent commercial entities: SpaceX's Starlink as the primary system and Amazon's Project Kuiper as the secondary. We saw that at virtually anytime, at least one secondary satellite has the potential to inflict prohibitively high interference onto a primary ground user, even with highly directional beams at both the user and satellite. However, it was also observed that there almost always exists one (or often more) secondary satellites which inflict acceptable interference onto the primary user while also delivering downlink SINR that approaches its upper bound. We showed that this is case even when the secondary system is not certain which primary satellite is serving that particular user. Based on these results, it can be concluded that in-band coexistence is indeed feasible through strategic satellite selection, but there remain open questions on practical mechanisms to execute such selection.

Interesting extensions of this evaluation on coexistence would be to cases where each satellite forms multiple beams, where the secondary system has uncertainty about the locations or receivers of primary ground users, and where the secondary system is tasked with scheduling satellites across its entire network. Creating novel techniques that leverage satellite selection or other means—potentially harnessing machine learning—to facilitate coexistence between LEO satellite communication systems on a network scale would be extremely valuable contributions. It would also be useful to formulate and extensively evaluate other interference protection constraints, considering, for example, a probabilistic constraint in time, constraints on handover frequency, and even user mobility. Investigating how coexistence may be facilitated by limited cooperation between the primary and secondary systems would make for interesting future work as well.

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